

Archive Memory as a Reconstructible Recovery Code in Deep Quality-Diversity Search

Takayuki Hasui – Independent Researcher

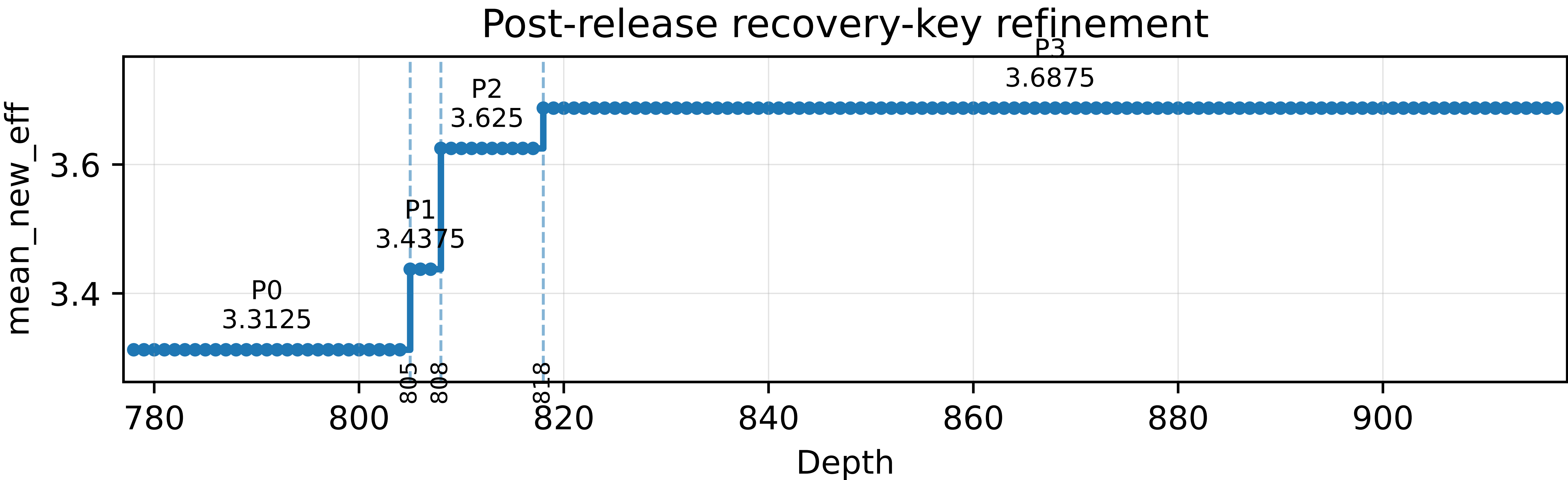
A Quality-Diversity archive can support reconstruction of a high-efficiency recovery key after ordinary continuation fails, with evidence pointing to a sparse, neighborhood-specific archive substrate rather than exact stored-row replay.

1. Can an archive do more than preserve diversity?

- Quality-Diversity (QD) archives are usually interpreted as diversity reservoirs.
- In deep sequential QCEP search, ordinary continuation can reach states where none of the tested candidates produces a stable successor.
- Archive-conditioned recovery can nevertheless restore high-efficiency structure.

Can archive memory function as a reconstructible recovery code?

2. Evidence 1: Persistent refinement



After archive-pool release, the selected recovery key refined through four phase-like regimes: P0 → P1 → P2 → P3. Efficiency increased from 3.3125 to 3.6875. P3 persisted across depths 818–917, giving an observed lower bound of at least 100 consecutive depths.

P3: 3.6875 for ≥ 100 depths

3. Evidence 2: Reconstructible, not replayed

P1, P2, and P3 recovery keys first appeared at their respective transition depths, rather than being exposed as old stored entries.

More directly, removing all stored P3-phase rows did not eliminate P3. Archive-conditioned recovery regenerated the same P3 target at the same efficiency of 3.6875.

Exact stored-row replay is insufficient to explain P3 recovery.

4. Evidence 3: Local substrate matters

Remove P3 neighborhood → P3 lost; fallback to P2
3.6875 → 3.6250

Rows retained	P3 neighborhood	Non-neighborhood
1	5/6 (83%)	0/6 (0%)
2	6/6 (100%)	0/6 (0%)

Equal-sized non-neighborhood add-backs did not restore P3. This supports a neighborhood-specific reconstructive substrate rather than recovery caused by adding arbitrary archive rows.

Archive-conditioned recovery:

P3 maintained at 3.6875 across checked depths 828–837.

Ordinary continuation:

No successful stable successor in the corresponding P3-phase continuation diagnostics.

5. Interpretation: Reconstructible archive basin

Together, these diagnostics support interpreting P3 as a reconstructible archive basin. Recovery information appears to be distributed across local archive structure rather than stored in one isolated row. In this controlled system, the archive behaves as an adaptive memory substrate for recovery after local search failure.

6. Limitations

- Controlled single-system study
- One graph topology and one depth-indexed QD regime
- Generalization across seeds, graph families, and other QD domains remains open

We do not claim a universal learning mechanism or a universal attractor.



Takayuki Hasui — Independent Researcher
Contact: ktoqq24470@yahoo.co.jp
Landing page (QR): GitHub public mirror poster page

ALIFE 2026
Poster 1b101s1